

Session Control in FFN weights

The Routing Problem

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Problem & Solution

- If alphaEdit correctly stores context for each session, what other parameters are necessary for routing to the correct session when a new read session occurs?
- E.g. "I have a guitar" and "I have a piano" is almost identical and challenging to route to a proper session

Skim connects dialogue to parameter edits

WRITE THE CURRENT TURN

Dialogue
turn

Select spans
Generate questions

Optimize residuals
Solve FFN update

$\Delta W(s)$

READ MEMORY LATER

Later question

Model with $W(\text{base}) + \Delta W(s)$

Answer

Research Hypothesis

Thesis statement

Test whether Llama 3.1 can recall selected session facts from FFN deltas while isolating other sessions and preserving existing behavior.

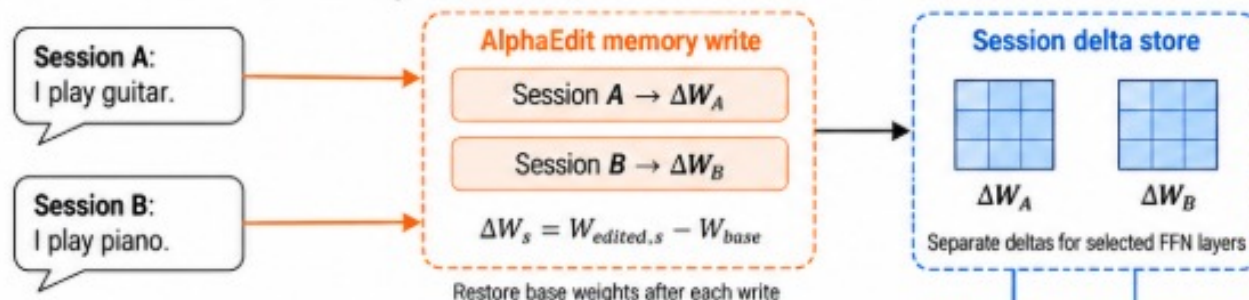
Testable hypotheses

- H2 Routing to appropriate sessions during session-switching.
- H3 Null-space projection preserves base behavior during continual writes.
- H4 Recall stays usable as sessions and stored turns increase.

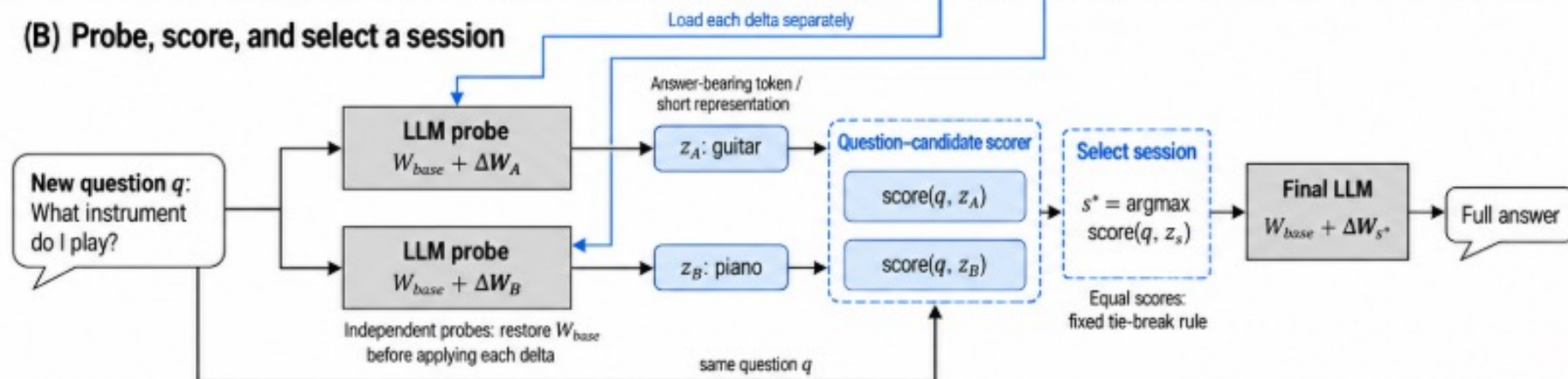
Session routing by delta probing

Proposed architecture

(A) Write and store session-specific deltas



(B) Probe, score, and select a session



Compatibility scores rank candidate answers; equally compatible answers may not identify the intended session.

Orange: AlphaEdit write Blue: session memory and routing Gray: LLM computation

Thank you

